Hochschild Homology of the b-Pseudodifferential Symbol Algebra

Changwei Zhou

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Abstract

Let X be a compact n-manifold with embedded boundary and let $E \to X$ be a complex vector bundle. We compute the continuous Hochschild homology of the Laurent complete-symbol algebra of classical b-pseudodifferential operators:

$$\mathcal{A}_{b,L}(X;E) = \frac{\displaystyle\bigcup_{m\in\mathbb{Z},\,\ell\geq 0} x^{-\ell}\Psi_b^m(X;E)}{\displaystyle\bigcup_{\ell\geq 0} x^{-\ell}\Psi_b^{-\infty}(X;E)}.$$

Here x is a boundary defining function. With the natural inductive-limit topology and completed Hochschild chains,

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}({}^bS^*X \times S^1_\sigma; \mathbb{C}).$$

The circle S^1_σ records symbol order and H_L^* is Laurent de Rham cohomology at the boundary. We derive the formula through the order filtration, the Hochschild–Kostant–Rosenberg map, the Poisson differential, and symplectic duality. We also separate this result from the Hochschild homology of an unreduced operator algebra, which depends on the chosen residual ideal and completion.

1 The precise statement

The phrase *the algebra generated by b-pseudodifferential operators* is ambiguous. At least four choices affect Hochschild homology: the allowed operator orders, the allowed powers of a boundary defining function, the residual ideal, and the locally convex completion. We use the standard algebra for which the homology has a finite-dimensional geometric description. It is the b-groupoid specialization of the Laurent symbol algebras studied in [7, 2, 1].

Let $x \in C^\infty(X)$ satisfy $x \geq 0$, $\partial X = \{x = 0\}$, and $dx \neq 0$ on ∂X. Put

$$\Psi_{b,L}^{\mathbb{Z}}(X;E) = \bigcup_{m\in\mathbb{Z},\,\ell\geq 0} x^{-\ell}\Psi_b^m(X;E), \qquad \Psi_{b,L}^{-\infty}(X;E) = \bigcup_{\ell\geq 0} x^{-\ell}\Psi_b^{-\infty}(X;E).$$

Both are given their natural strict inductive-limit topologies. The second space is a two-sided ideal in the first.

Definition 1.1. The *Laurent complete-symbol algebra* is

$$\mathcal{A}_{b,L}(X;E) = \Psi_{b,L}^{\mathbb{Z}}(X;E)/\Psi_{b,L}^{-\infty}(X;E).$$

Its continuous Hochschild chains are

$$C_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) = \mathcal{A}_{b,L}^{\widehat{\otimes}(k+1)},$$

with the completed projective tensor product and the usual Hochschild boundary b.

Write

$$Y = {}^bS^*X = ({}^bT^*X \setminus 0)/\mathbb{R}_+, \qquad Z = Y|_{\partial X}.$$

The boundary of Y is Z. The symbol-order circle is denoted S^1_σ.

Theorem 1.2 (Hochschild homology)**.** *For every $k \geq 0$,*

$$\boxed{\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}(Y \times S^1_\sigma;\mathbb{C}).} \tag{1}$$

The answer is independent of E. If X has one embedded boundary hypersurface, then

$$\begin{aligned}\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H^{2n-k}(Y) \oplus H^{2n-k-1}(Y) \\ \oplus H^{2n-k-1}(Z) \oplus H^{2n-k-2}(Z),\end{aligned} \tag{2}$$

where a cohomology group in negative degree is zero. In particular the homology vanishes for $k > 2n$.

The isomorphism in (1) is the intrinsic statement. The four summands in (2) use the Laurent face decomposition and the Künneth splitting for S^1_σ.

$$\begin{tikzcd}
\mathcal{A}_{b,L} \arrow[r, "\text{order filtration}"] & \mathrm{gr}\,\mathcal{A}_{b,L} \arrow[r, "\text{HKR}"] & \Omega_L^*({}^bT^*X \setminus 0) \arrow[d, "\delta_{\pi_b}"] \\
\mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L}) & H_L^{2n-*}(Y \times S^1_\sigma) \arrow[l, "E^2=E^\infty"'] & H_*^{\pi_b,L}({}^bT^*X \setminus 0) \arrow[l, "\text{symplectic duality}"']
\end{tikzcd}$$

Figure 1: The computation passes from the filtered noncommutative algebra to its commutative associated graded, then from Poisson homology to Laurent de Rham cohomology.

2 The b-cotangent geometry

The Lie algebra of b-vector fields is

$$\mathcal{V}_b(X) = \{V \in C^\infty(X;TX) : V \text{ is tangent to } \partial X\}.$$

It is the space of smooth sections of a vector bundle ${}^bTX \to X$. Near a boundary point, choose product coordinates

$$(x, y_1, \ldots, y_{n-1}).$$

Then

$$\mathcal{V}_b(X) = C^\infty(X)\text{-span}\{x\partial_x, \partial_{y_1}, \ldots, \partial_{y_{n-1}}\}.$$

The dual local frame of ${}^bT^*X$ is

$$\frac{dx}{x}, \quad dy_1, \ldots, dy_{n-1}.$$

Thus a b-covector has the form

$$\xi\frac{dx}{x} + \eta \cdot dy.$$

On ${}^bT^*X$ the tautological b-one-form and its differential are

$$\alpha_b = \xi\frac{dx}{x} + \eta \cdot dy, \qquad \omega_b = d\alpha_b = d\xi \wedge \frac{dx}{x} + d\eta \wedge dy.$$

The form ω_b is nondegenerate as a b-two-form. Its inverse is the b-Poisson tensor

$$\pi_b = x\partial_x \wedge \partial_\xi + \sum_j \partial_{y_j} \wedge \partial_{\eta_j}.$$

This is the geometric bracket appearing in the first nonzero differential of the Hochschild spectral sequence.

3 The associated graded algebra

Filter $\mathcal{A}_{b,L}$ by operator order:

$$F_m \mathcal{A}_{b,L} = \text{image}\left(\bigcup_{\ell \geq 0} x^{-\ell} \Psi_b^m(X;E)\right), \qquad m \in \mathbb{Z}.$$

The filtration is exhaustive, separated in the complete-symbol sense, and multiplicative. Principal symbols give

$$\mathrm{gr}_m \mathcal{A}_{b,L} \cong \mathcal{O}_L({}^bT^*X \setminus 0; \mathrm{End}\, E)_m,$$

the Laurent functions that are homogeneous of degree m in the fibers.

Lemma 3.1 (Morita reduction)**.** *The coefficient bundle E does not change the Hochschild homology.*

Proof. Locally, the associated graded algebra with coefficients is a matrix algebra over the scalar symbol algebra. The local Morita maps are compatible with changes of frame, and the standard trace and matrix-unit homotopies patch by a partition of unity. Hence the completed Hochschild complexes for scalar symbols and for $\mathrm{End}\, E$-valued symbols are quasi-isomorphic. The same filtered argument lifts the conclusion from the associated graded algebra to $\mathcal{A}_{b,L}(X;E)$. □

We therefore suppress E. The Hochschild–Kostant–Rosenberg map on the associated graded algebra is

$$\chi(a_0 \otimes \cdots \otimes a_j) = \frac{1}{j!} a_0\, da_1 \wedge \cdots \wedge da_j.$$

Proposition 3.2 (The E^1-page)**.** *The HKR map identifies the first page of the order spectral sequence with Laurent differential forms on ${}^bT^*X \setminus 0$, decomposed by fiber homogeneity.*

Proof. On each b-cotangent chart the associated graded product is ordinary multiplication of scalar homogeneous symbols. The continuous HKR theorem identifies its Hochschild homology with differential forms. Laurent poles in x cause no change in the local contracting homotopy: every chain has a uniform finite pole order, and differentiation preserves the Laurent class. The construction is local and commutes with restriction, so a partition of unity patches the chartwise identifications. Fiber homogeneity is preserved throughout. □

4 The first differential is Poisson homology

For classical b-symbols a and c, symbolic composition gives

$$\sigma_{m+m'-1}([A,C]) = \frac{1}{i}\{a,c\}_b.$$

Consequently the first differential after HKR is, up to the harmless factor i, the Poisson differential

$$\delta_b = [\iota_{\pi_b}, d] = \iota_{\pi_b} d - d\iota_{\pi_b}.$$

Let $N = 2n$ be the dimension of ${}^bT^*X$. Symplectic duality is the map

$$*_b : \Omega_L^j({}^bT^*X \setminus 0) \longrightarrow \Omega_L^{N-j}({}^bT^*X \setminus 0)$$

defined by contraction with the symplectic volume $\omega_b^n/n!$. In Darboux b-coordinates, the standard calculation gives

$$*_b\delta_b = (-1)^{j+1} d *_b \quad \text{on } \Omega_L^j. \tag{3}$$

It remains valid at $x=0$, because both sides use the b-frame $(dx/x, dy, d\xi, d\eta)$ and preserve finite Laurent pole order.

Corollary 4.1. *The Poisson homology on the E^2-page is Laurent de Rham cohomology with complementary degree:*

$$H_j^{\pi_b,L}({}^bT^*X \setminus 0) \cong H_L^{2n-j}({}^bT^*X \setminus 0).$$

5 Homogeneity and the symbol-order circle

Choose a positive fiber norm r on ${}^bT^*X \setminus 0$. Then

$${}^bT^*X \setminus 0 \cong Y \times \mathbb{R}_+, \qquad (q,r) \longmapsto rq.$$

Only total homogeneity zero survives in the de Rham cohomology of the formal homogeneous expansion. Indeed, if $\mathcal{E} = r\partial_r$ is the Euler vector field and a form α has nonzero homogeneity w, Cartan's formula gives

$$(d\iota_{\mathcal{E}} + \iota_{\mathcal{E}} d)\alpha = \mathcal{L}_{\mathcal{E}}\alpha = w\alpha.$$

Thus $w^{-1}\iota_{\mathcal{E}}$ contracts every nonzero weight.

At weight zero, a form splits uniquely as

$$\alpha(r,q) = \alpha_0(q) + \frac{dr}{r} \wedge \alpha_1(q).$$

The formal order variable dr/r is represented cohomologically by the generator $d\theta_\sigma$ of a circle. Hence

$$H_L^j({}^bT^*X \setminus 0)_{\text{hom}=0} \cong H_L^j(Y \times S^1_\sigma). \tag{4}$$

This is the source of the extra S^1 in the answer; it is not a geometric circle in the fibers of ${}^bT^*X$.

6 Laurent cohomology and the boundary face

Let M be a compact manifold with one embedded boundary hypersurface and defining function x. The Laurent de Rham complex is

$$\Omega_L^j(M) = \bigcup_{\ell \geq 0} x^{-\ell}\Omega^j(M), \qquad d : \Omega_L^j(M) \to \Omega_L^{j+1}(M).$$

Proposition 6.1 (Face decomposition)**.** *There is an isomorphism*

$$H_L^j(M) \cong H^j(M) \oplus H^{j-1}(\partial M).$$

The boundary summand is represented by $\widetilde{\beta} \wedge d(\chi \log x)$, where β is a closed $(j-1)$-form on ∂M, $\widetilde{\beta}$ is a collar extension, and χ is a cutoff supported in the collar and equal to one near the boundary.

Proof. For germs at a boundary point, Taylor expansion in x reduces a Laurent form to a finite sum of powers x^q. Every term with $q \neq 0$ is exact in the normal direction because

$$x^{q-1}dx = \frac{1}{q}d(x^q).$$

The two uncontracted normal classes are 1 and $dx/x = d\log x$. Thus the cohomology sheaf of the Laurent complex has one copy of the constant sheaf in degree zero and one boundary-supported copy in degree one. The ordinary de Rham resolution in the tangential variables identifies their global cohomology with $H^j(M)$ and $H^{j-1}(\partial M)$, respectively. The displayed cutoff construction realizes the boundary class globally. $\square$

Apply Proposition 6.1 to $M = Y \times S^1_\sigma$, whose boundary is $Z \times S^1_\sigma$. Then use

$$H^r(W \times S^1) \cong H^r(W) \oplus H^{r-1}(W).$$

For $r = 2n - k$ this gives exactly the four summands in (2).

$$\begin{tikzcd}
H^r_L(Y \times S^1_\sigma) \arrow[d, equal] & & r = 2n-k \\
H^r(Y \times S^1_\sigma) \oplus H^{r-1}(Z \times S^1_\sigma) \arrow[d, equal] & & \\
\big(H^r(Y) \oplus H^{r-1}(Y)\big) \oplus \big(H^{r-1}(Z) \oplus H^{r-2}(Z)\big) & &
\end{tikzcd}$$

Figure 2: The four summands come from the ordinary class, the symbol-order logarithm $d\log r$, the boundary logarithm $d\log x$, and their product.

For a manifold with corners, the same local argument has one generator $d\log x_H$ for every boundary hypersurface H through a point. If $\mathcal{F}_j(M)$ denotes the codimension-j faces, then

$$H^r_L(M) \cong \bigoplus_{j \geq 0} \bigoplus_{F \in \mathcal{F}_j(M)} H^{r-j}(F). \tag{5}$$

7 Collapse and assembly of the computation

Combining HKR, the Poisson differential, symplectic duality, and homogeneity gives

$$E^2_k \cong H^{2n-k}_L(Y \times S^1_\sigma).$$

The remaining input is the collapse theorem for Laurent complete symbols.

Theorem 7.1 (Benameur–Nistor collapse theorem)**.** *For the order-filtered continuous Hochschild complex of a Laurent complete-symbol algebra associated with a b-groupoid, the spectral sequence converges and degenerates at E^2.*

This is the point at which we use the convergence and degeneration theorem of [1]; the preceding identifications make its E^2-page explicit in the present b-setting.

This theorem is proved by first working over product charts, where the filtered algebra is a completed formal symbol algebra, and then using the locality of continuous Hochschild homology to glue the calculation. The Laurent filtration supplies uniform bounds on pole orders, and the order filtration supplies completeness in the negative direction. These are the two points needed for convergence; without them the formal E^2-calculation need not determine the homology of the original algebra.

The theorem identifies E^2 with the associated graded of Hochschild homology. A quantization map splits the order filtration as a topological vector space, giving (1). Different quantizations change the splitting but not the intrinsic H_L^{2n-k} description. This completes the computation.

8 Checks and consequences

8.1 The closed-manifold case

If $\partial X = \varnothing$, Laurent cohomology is ordinary cohomology and $Z = \varnothing$. Formula (2) reduces to

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}(X;E)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X),$$

equivalently $H^{2n-k}(S^*X \times S^1_\sigma)$. This is the Brylinski–Getzler computation [3].

8.2 Degree zero and residue traces

Since $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}/[\mathcal{A},\mathcal{A}]$, continuous traces factor through HH_0. Theorem 1.2 gives

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H_L^{2n}(Y \times S^1_\sigma).$$

If $n \geq 2$, X is connected, and ∂X is connected, then Y has nonempty boundary and $Z \to \partial X$ has connected sphere fiber. Dimension considerations reduce the right side to $H^{2n-2}(Z) \cong \mathbb{C}$. Thus the Laurent complete-symbol algebra has one residue trace up to scale. With several boundary components, one obtains a boundary-face residue for each connected component of Z.

8.3 A product cylinder

Let $X = [0,1] \times N$, where N is closed of dimension $n-1$, and put

$$W = S(\mathbb{R} \oplus T^*N).$$

Then $Y \simeq [0,1] \times W$ and $Z = W \sqcup W$. Therefore

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-k}(W) \oplus H^{2n-k-1}(W)^{\oplus 3} \oplus H^{2n-k-2}(W)^{\oplus 2}.$$

This example separates the order-circle contribution from the two boundary faces.

9 What the theorem does not compute

The unreduced operator algebra fits into

$$0 \longrightarrow \Psi_{b,L}^{-\infty}(X;E) \longrightarrow \Psi_{b,L}^{\mathbb{Z}}(X;E) \longrightarrow \mathcal{A}_{b,L}(X;E) \longrightarrow 0. \tag{6}$$

The answer for $\Psi^{\mathbb{Z}}_{b,L}$ is not obtained by simply reusing (1). It depends on the topology placed on the residual ideal and on whether kernels are required to vanish rapidly at the side faces, merely have polyhomogeneous expansions, or are completed in an operator norm.

For an H-unital residual ideal, excision gives the long exact sequence

$$\cdots \to \mathrm{HH}_k(\Psi^{-\infty}_{b,L}) \to \mathrm{HH}_k(\Psi^{\mathbb{Z}}_{b,L}) \to \mathrm{HH}_k(\mathcal{A}_{b,L}) \xrightarrow{\partial} \mathrm{HH}_{k-1}(\Psi^{-\infty}_{b,L}) \to \cdots .$$

Melrose and Nistor compute the analogous two-filtration sequence in the cusp calculus and explain how the same mechanism applies to the b-calculus [7]. The connecting maps are the algebraic location of the index and eta corrections. Thus (1) is the complete answer for Laurent complete symbols, while (6) is the correct starting point for any specified unreduced operator algebra.

10 Relation to index theory

An elliptic b-operator determines an invertible matrix over the complete-symbol algebra. Its K_1-class pairs with cyclic cocycles obtained from the Hochschild classes above. The ordinary symbol residue detects the interior term; the boundary logarithmic class detects the indicial or eta term. In the operator extension, the boundary map sends the symbol class to the Fredholm index whenever the normal operator is invertible.

The work of Liu and Ma on differential K-theory and eta-invariant localization supplies a geometric refinement of this pairing: differential forms and eta data refine the topological K-class. It does not replace the Hochschild spectral-sequence calculation [4, 5]. The logical division is therefore

Melrose–Nistor and Benameur–Nistor: algebraic homology,

Liu–Ma: differential-K and eta localization of index data.

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